Question	Points	Correct	Incorrect	Time (hh:mm:ss)	Results

FILTERS: Show All Categories ▾ Filter Questions

- OS disk type: Standard HDD
- Ultra Disk compatibility: Disabled
- Managed disks: Disabled

To prevent downtime, you need to make sure that the instance can be moved in different Availability Zones using Site Recovery.

Which parameter should be modified?

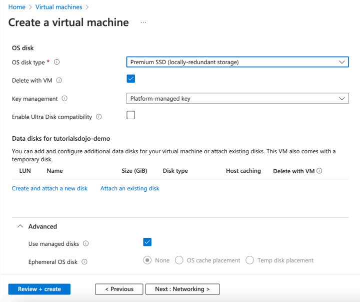
(view) 1 0 1 00:00:00

○ Ultra Disk compatibility
○ OS disk type
● Managed disks
○ Region

Azure Site Recovery helps ensure business continuity by keeping business apps and workloads running during outages. Site Recovery replicates workloads running on physical and virtual machines (VMs) from a primary site to a secondary location. When an outage occurs at your primary site, you fail over to a secondary location, and access apps from there. After the primary location is running again, you can fail back to it.

Site Recovery can manage replication for:

- Azure VMs replicating between Azure regions
- Replication from Azure Public Multi-Access Edge Compute (MEC) to the region
- Replication between two Azure Public MECs
- On-premises VMs, Azure Stack VMs, and physical servers


Home > Virtual machines >

Create a virtual machine ...

OS disk

OS disk type *: Premium SSD (locally-redundant storage)

Delete with VM: ☑

Key management: Platform-managed key

Enable Ultra Disk compatibility: ☐

Data disks for tutorialsdojo-demo

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM

Create and attach a new disk   Attach an existing disk

Advanced

Use managed disks: ☑

Ephemeral OS disk: ● None ○ OS cache placement ○ Temp disk placement

Managed disks are designed to have a 99.999% uptime. Managed disks achieve this by storing three copies of your data, resulting in high durability. If one or two replicas fail, the remaining replicas help ensure data persistence and high failure tolerance.

With Azure Site Recovery, you can move single-instance VMs into Availability Zones in a target region. However, in order to move a VM to an Availability Zone, you must first ensure that the VM is using managed disks. You can also convert existing Windows VMs that use unmanaged disks to use managed disks.

Hence, the correct answer is: **Managed disks.**

Region is incorrect because South Central US already allows you to select Availability Zone as an option. This means that you can move the VM to different AZs. Take note that some Regions or locations don't support AZs.

OS disk type is incorrect because it doesn't matter what type of disk the VM is using, if you already enabled "use managed disks" in advanced disk configuration.

Ultra Disk compatibility is incorrect because the requirement is not related to data-intensive workloads. You only need to ensure that the VMs can be moved to a different AZ in the event of a disaster.

References:

https://learn.microsoft.com/en-us/azure/site-recovery/move-azure-vms-avset-azone

https://learn.microsoft.com/en-us/azure/virtual-machines/windows/convert-unmanaged-to-managed-disks

Check out this Azure Virtual Machines Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-machines/

8 You are managing an Azure subscription that contains the following resources:

Name	Type	Resource group	Location
TD-RG1	Resource group	N/A	Australia East
TD-RG2	Resource group	N/A	Southeast Asia
TD-RG3	Resource group	N/A	East Asia
TD-VM1	Virtual machine	TD-RG1	Australia East
TD-VMSS1	Virtual machine scale set	TD-RG3	Australia East
TD-Proximity1	Proximity placement group	TD-RG2	Southeast Asia
TD-Proximity2	Proximity placement group	TD-RG3	East Asia
TD-Proximity3	Proximity placement group	TD-RG1	Australia East

You plan to configure a proximity placement group for the `TD-VMSS1` virtual machine scale set.

Which of the following proximity placement groups should you use?

(view) 1 0 1 00:00:00

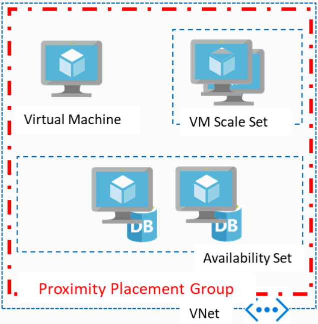
○ `TD-Proximity3`

● `TD-Proximity1` , `TD-Proximity2` , and `TD-Proximity3`

○ `TD-Proximity2`

○ `TD-Proximity1` and `TD-Proximity3`

Azure virtual machine scale sets let you create and manage a group of load-balanced VMs. The number of VM instances can automatically increase or decrease in response to demand or a defined schedule. Scale sets provide high availability to your applications and allow you to centrally manage, configure, and update a large number of VMs. With virtual machine scale sets, you can build large-scale services for areas such as compute, big data, and container workloads.



A proximity placement group is a logical grouping used to make sure that Azure compute resources are physically located close to each other. Proximity placement groups are useful for workloads where low latency is a requirement. When you assign your virtual machines to a proximity placement group, the virtual machines are placed in the same data center, resulting in lower and deterministic latency for your applications.

It is stated in the scenario that you must configure a placement group for `TD-VMSS1` . Among the given placement groups, you can only assign `TD-Proximity3` since it belongs to the same region as `TD-VMSS1` . Remember that when you are configuring a proximity placement group for a virtual machine scale set. Both the placement group and scale set must be in the same region.

Hence, the correct answer is: **`TD-Proximity3`** .

The option that says: **`TD-Proximity2`** is incorrect because `TD-VMSS1` is located in East Asia and not in Australia East. Although both `TD-VMSS1` and `TD-Proximity2` belong to the same resource group, take note that the location of the resource group is irrelevant in this scenario. You should assign `TD-VMSS1` to the `TD-Proximity3` placement group to satisfy the requirement.

The option that says: **`TD-Proximity1` and `TD-Proximity3`** is incorrect. You can't configure `TD-Proximity1` for `TD-VMSS1` since the location of `TD-Proximity1` is in Southeast Asia while `TD-VMSS1` is in Australia East. The region of the virtual machine scale set and the proximity placement group should be the same.

The option that says: **`TD-Proximity1` , `TD-Proximity2` , and `TD-Proximity3`** is incorrect because you can only assign `TD-VMSS1` in `TD-Proximity3` . In this scenario, both the virtual machine scale set and proximity placement group must be in the same region.

References:

https://docs.microsoft.com/en-us/azure/virtual-machines/windows/proximity-placement-groups-portal

https://azure.microsoft.com/en-us/blog/announcing-the-general-availability-of-proximity-placement-groups/

Check out this Azure Virtual Machines Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-machines/

10 You plan to automate the deployment of Windows Servers using a virtual machine scale set.

You need to make sure that the web components are installed in the virtual machines.

Which actions should you perform? (Select TWO.)

(view) 1 0 1 00:00:00

☐ Create a configuration script.

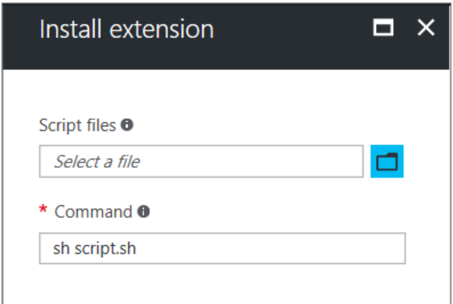
☐ Create an automation account.

☑ Download and install the VPN client configuration package.

☐ Configure the `extensionProfile` section of the ARM template.

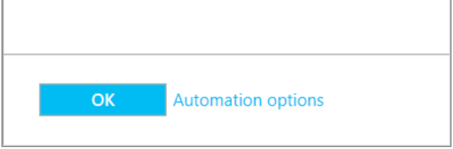
☑ Create a new scale set.

Azure virtual machine scale sets let you create and manage a group of load-balanced VMs. The number of VM instances can automatically increase or decrease in response to demand or a defined schedule. Scale sets provide high availability to your applications and allow you to centrally manage, configure, and update a large number of VMs.



* Command

sh script.sh



The Custom Script Extension downloads and executes scripts on Azure virtual machines. This extension is useful for post-deployment configuration, software installation, or any other configuration or management tasks.

Hence, the correct answers are:

- **Create a configuration script.**
- **Configure the `extensionProfile` section of the ARM template.**

The option that says: **Create an automation account** is incorrect because an automation account wouldn't help you automatically install web components. You still need to create a configuration script and `extensionProfile` in the ARM template.

The option that says: **Download and install the VPN client configuration package** is incorrect because it is primarily used to configure Point-to-Site VPN connectivity between a client machine and an Azure Virtual Network; it has nothing to do with installing web components on virtual machines in a scale set.

The option that says: **Create a new scale set** is incorrect because it wouldn't install the required web components. Instead of creating a new scale set, you just need to use a custom script extension to install the web components in the VMs.

References:

https://docs.microsoft.com/en-us/azure/virtual-machine-scale-sets/virtual-machine-scale-sets-deploy-app

https://docs.microsoft.com/en-us/azure/virtual-machine-scale-sets/tutorial-install-apps-template#what-is-the-azure-custom-script-extension

https://docs.microsoft.com/en-us/azure/virtual-machine-scale-sets/virtual-machine-scale-sets-deploy-app#already-provisioned

Check out this Azure Virtual Machines Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-machines/

11 You are tasked with setting up an automated deployment pipeline in Azure DevOps for a web application.

`WebAppRG` is the resource group that is part of your Azure subscription.

You plan to deploy a web app named `MyApp` .

`Webapp.bicep` is the name of the Bicep file you have.

The company already uses service endpoints to restrict network access to its Azure Storage accounts.

You should modify `webapp.bicep` to enable the automated deployment of `MyApp` to `WebAppRG` .

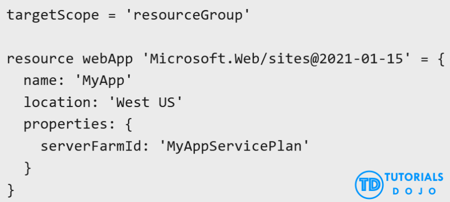
Which of the following properties should be modified?

(view) 1 0 1 00:00:00

- ● `scope`
- ○ `targetScope`
- ● `location`
- ○ `tags`

In Azure, resources are organized hierarchically into management groups, subscriptions, resource groups, and resources. This hierarchy allows you to manage and organize your resources based on your business needs. The `scope` property in a Bicep file is used to specify the level in this hierarchy where your resources are deployed.

When you're setting up an automated deployment pipeline for a web application like `MyApp` , you need to specify where in your Azure environment the app should be deployed. This is where the `scope` property comes into play.



```
targetScope = 'resourceGroup'

resource webApp 'Microsoft.Web/sites@2021-01-15' = {
  name: 'MyApp'
  location: 'West US'
  properties: {
    serverFarmId: 'MyAppServicePlan'
  }
}
```

The `scope` property in a Bicep file is used to specify the scope at which the resource is deployed. In the given question, you would modify the `scope` property to point to the `WebAppRG` resource group in your Azure subscription. This would enable the automated deployment of `MyApp` to `WebAppRG` .

Hence, the correct answer is: **`scope.`**

`targetScope` is incorrect because this property defines the type of deployment scope at the file level. It tells Azure whether the Bicep file deploys to a resource group, subscription, or management group, but it does not specify which specific resource group to deploy to.

`location` is incorrect because the `location` property is used to specify the geographic location where the resource is hosted. While it's an important property for many resources, modifying it would not change the deployment scope of the resource.

`tags` is incorrect because `tags` are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Modifying `tags` would not affect the deployment scope of the resource.

References:

https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/bicep-functions-scope

https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/file

Check out this Azure Cheat Sheet:

https://tutorialsdojo.com/microsoft-azure-cheat-sheets/

12 A startup has an Azure subscription that contains the following resources:

Name	Type
tdvm	Virtual Machine
tdsql	SQL database
tdvnet	Virtual Network
tdsa1	Storage Account

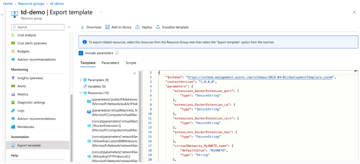
You have been tasked with replicating the current state of your resources in order to automate future deployments when a new feature needs to be added to the application.

Which of the following should you do?

(view) 1 0 1 00:00:00

○ Create a VM with preset configurations.
○ Redeploy and reapply a VM.
○ Use the resource group export template.
○ Capture an image of a VM.

Azure Resource Manager (ARM) templates is a service provided by Microsoft Azure that allows you to provision, manage, and delete Azure resources using declarative syntax. These templates can be used to deploy and manage resources such as virtual machines, storage accounts, and virtual networks in a consistent and reliable manner. To deploy the template, you can use the Azure Portal, Azure CLI, or Azure PowerShell.



To export related resources, select the resources from the Resource Group view then select the "Export template" option from the tool bar.

```
{
    "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
    "contentVersion": "1.0.0.0",
    "parameters": {
        "extensions_DockerExtension_port": {
            "type": "SecureString"
        },
        "extensions_DockerExtension_ca": {
            "type": "SecureString"
        },
        "extensions_DockerExtension_cert": {
            "type": "SecureString"
        },
        "extensions_DockerExtension_key": {
            "type": "SecureString"
        },
        "virtualNetworks_MyVNETD_name": {
            "defaultValue": "MyVNETD",
            "type": "String"
        },
```

In this scenario, you need to use ARM export templates to replicate the current state of our resources. This means that if you need to redeploy all your resources, you can just create a reusable template instead of going all over the manual creation of resources. You can export a service or a resource group. Based on the given requirements, we just need to capture all resources, then export resource group as template.

Hence, the correct answer is: **Use the resource group export template.**

The option that says: **Capture an image of a VM** is incorrect because this just creates a snapshot of the virtual machine configurations. Take note that you need to capture the current state of all resources. Therefore, export template will help us ease the creation of resources.

The option that says: **Redeploy and reapply a VM** is incorrect because redeploying the VM just migrates it to a new Azure host. While reapply is used to resolve the issues of a stuck failed state of the VM. Both features does not help us capture the current state of our resources.

The option that says: **Create a VM with preset configurations** is incorrect because this only helps you choose a VM based on your workload type and environment.

References:

https://learn.microsoft.com/en-us/azure/azure-resource-manager/templates/export-template-portal

https://learn.microsoft.com/en-us/azure/azure-resource-manager/templates/overview

Check out this Azure Resource Manager (ARM) Cheat Sheet:

https://tutorialsdojo.com/azure-resource-manager-arm/

Sub-Total: 12 7 (58.33%) 5 (41.67%) 00:00:00 58.33%

Category: **AZ-104 - Implement and Manage Storage**

3 You have an Azure blob storage account in your Azure subscription named `TD1`, located in the Southeast Asia region. The company already uses replication policies for other data duplication needs. Due to compliance requirements, data uploaded to `TD1` must be duplicated to the Australia Central region for redundancy. The solution must minimize administrative effort.

What should you do?

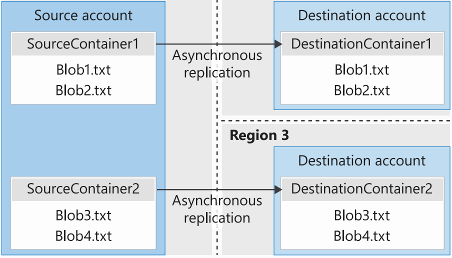
(view) 1 0 1 00:00:00

○ Configure object replication.
○ Configure firewalls and virtual networks.
○ Configure versioning.
○ Configure Geo-redundant storage (GRS).

Object replication asynchronously copies block blobs between a source storage account and a destination account. Some scenarios supported by object replication include:

1. Minimizing latency. Object replication can reduce latency for read requests by enabling clients to consume data from a region that is in closer physical proximity.
2. Increase efficiency for compute workloads. With object replication, compute workloads can process the same sets of block blobs in different regions.
3. Optimizing data distribution. You can process or analyze data in a single location and then replicate just the results to additional regions.
4. Optimizing costs. After your data has been replicated, you can reduce costs by moving it to the archive tier using life cycle management policies.

Region 1
Region 2



The requirement states that whenever data is uploaded to `TD1` must be duplicated to Australia Central due to compliance requirements. Since the regional pair of Southeast Asia is East Asia, we won't be able to use geo-redundant storage (GRS) as we cannot choose the secondary region due to regional pairs. Instead, we can use object replication to copy data from `TD1` to a storage account in the Australia Central region.

Object replication is supported for general-purpose v2 storage accounts and premium block blob accounts. Both the source and destination accounts must be either general-purpose v2 or premium block blob accounts. Object replication supports block blobs only; append blobs and page blobs aren't supported.

Hence, the correct answer is: **Configure object replication.**

Configure firewalls and virtual networks is incorrect because this feature only allows users of Azure storage accounts to block or allow specific traffic to your storage account. It does not have any capability to replicate data to another region.

Configure versioning is incorrect because this allows you to automatically maintain previous versions of an object in a single storage account. Although to use object replication, versioning must be enabled in the source and target storage accounts.

Configure Geo-redundant storage (GRS) is incorrect because the data will automatically be stored in East Asia since it is the regional pair of the Southeast Asia region. You don't get to choose the secondary region when enabling geo-redundant storage. Instead, use object replication.

References:

https://learn.microsoft.com/en-us/azure/storage/blobs/object-replication-overview

https://learn.microsoft.com/en-us/azure/reliability/cross-region-replication-azure

Check out this Azure Storage Overview Cheat Sheet:

https://tutorialsdojo.com/azure-storage-overview/

4 Note: This item is part of a series of case study questions with the exact same scenario but has a different technical requirement. Each one in the series has a unique solution that may or may not comply with the requirements specified in the scenario.

- Overview
- Existing Environment
- Planned Changes
- Technical Requirements

Tutorials Dojo is an online learning portal for technology-related topics that empowers its users to upgrade their skills and career. Tutorials Dojo has users worldwide, ranging from the United States, Europe, and Asia.

Question

What should you do?

(view) 1 0 1 00:00:00

○ Use Azure Import/Export service to copy the files.

○ Use file explorer to copy the files by mapping a drive using a shared access signature (SAS) in the Azure storage account to grant temporary access.

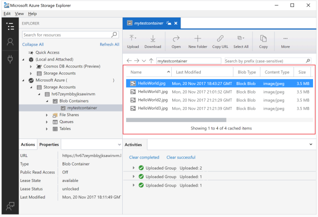
○ Use Azure Storage Explorer to copy the files.

○ Use file explorer to copy the files by mapping a drive using an Azure storage account access key for authorization.

Azure Blob storage is Microsoft's object storage solution for the cloud. Blob storage is optimized for storing massive amounts of unstructured data. Unstructured data is data that doesn't adhere to a particular data model or definition, such as text or binary data.

Blob storage is designed for:

- Serving images or documents directly to a browser.
- Storing files for distributed access.
- Streaming video and audio.
- Writing to log files.
- Storing data for backup and restore disaster recovery, and archiving.
- Storing data for analysis by an on-premises or Azure-hosted service.



Name	Last Modified	Blob Type	Content Type	Size
HelloWorld.jpg	Mon, 20 Nov 2017 18:43:27 GMT	Block Blob	image/jpeg	3.5 MB
HelloWorld1.jpg	Mon, 20 Nov 2017 21:01:32 GMT	Block Blob	image/jpeg	3.5 MB
HelloWorld2.jpg	Mon, 20 Nov 2017 21:21:29 GMT	Block Blob	image/jpeg	3.5 MB
HelloWorld3.jpg	Mon, 20 Nov 2017 21:21:39 GMT	Block Blob	image/jpeg	3.5 MB

Showing 1 to 4 of 4 cached items

URL	https://hr67zeymbbyjksawinvm.l
Type	Blob Container
Public Read Access	Off
Lease State	available
Lease Status	unlocked
Last Modified	Mon, 20 Nov 2017 18:11:49 GM

✓ Uploaded Group Uploaded: 2
✓ Uploaded Group Uploaded: 1
✓ Uploaded Group Uploaded: 1

Microsoft Azure Storage Explorer is a standalone app that is accessible, intuitive, feature-rich graphical user interface (GUI) for full management of cloud storage resources and makes it easy to work with Azure Storage data on Windows, macOS, and Linux. You can upload, download, and manage Azure blobs, files, queues, and tables, as well as Azure Cosmos DB and Azure Data Lake Storage entities.

The requirements to be considered for this scenario are:

- Migrate the media files to Azure over the Internet.
- The media files must be stored in a Blob container and cached via Content Delivery Network.

Hence, the correct answer is: **Use Azure Storage Explorer to copy the files.**

The option that says: **Use Azure Import/Export service to copy the files** is incorrect. Azure Import/Export service is primarily used to securely import large amounts of data to Azure Blob storage and Azure Files by shipping disk drives to an Azure datacenter. The requirement states that the transfer of the media files must be done over the Internet.

The following options are incorrect because you cannot mount a Blob container using file explorer. Take note that the requirement states that the media files must be stored in a Blob container.

- Use file explorer to copy the files by mapping a drive using a shared access signature (SAS) in the Azure storage account to grant temporary access.

- Use file explorer to copy the files by mapping a drive using an Azure storage account access key for authorization.

References:

https://azure.microsoft.com/en-us/features/storage-explorer/

https://docs.microsoft.com/en-us/azure/vs-azure-tools-storage-manage-with-storage-explorer

Check out this Azure Blob Storage Cheat Sheet:

https://tutorialsdojo.com/azure-blob-storage/

5 You have an Azure subscription named `Davao-Subscription` that contains an Azure Files named `Baguio-Share` .

You have several Azure virtual machines that are domain-joined to an on-premises Active Directory Domain Services (AD DS) domain. The domain controller has a healthy NetLogon service, and a site-to-site VPN connection provides cross-premises connectivity.

There is a requirement to replace your on-premises file server with `Baguio-Share` . Your domain-joined machines must be able to mount `Baguio-Share` using your AD DS credentials.

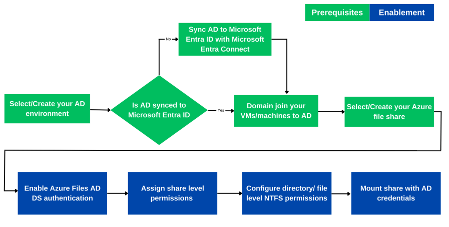
Which four actions should you perform in sequence?

Instructions: To answer, drag the appropriate item from the column on the left to its description on the right. Each correct match is worth one point.

(view) 4 0 4 00:00:00

Enable AD DS authentication	1
Assign share and directory permissions	2
Mount file share with AD credentials	3
Sync on-premises AD with Microsoft Entra Connect	4

Azure Files offers fully managed file shares in the cloud that are accessible via the industry standard Server Message Block (SMB) protocol or Network File System (NFS) protocol. Azure Files SMB file shares are accessible from Windows, Linux, and macOS clients. Azure Files NFS file shares are accessible from Linux or macOS clients. Additionally, Azure Files SMB file shares can be cached on Windows Servers with Azure File Sync for fast access near where the data is being used.


Prerequisites
Enablement
Sync AD to Microsoft Entra ID with Microsoft Entra Connect
No
Select/Create your AD environment
Is AD synced to Microsoft Entra ID
Yes
Domain join your VMs/machines to AD
Select/Create your Azure file share
Enable Azure Files AD DS authentication
Assign share level permissions
Configure directory/ file level NTFS permissions
Mount share with AD credentials

Enabling AD DS authentication for your Azure file shares allows you to authenticate to your Azure file shares with your on-premises credentials. Further, it allows you to better manage your permissions to allow granular access control.

To enable AD DS authentication, you must do the following in sequence:

1. Sync on-premises AD with Microsoft Entra Connect

- Identities used for access must be synced to Microsoft Entra ID. Only hybrid users that exist in both on-premises AD DS and Microsoft Entra ID can be authenticated and authorized for Azure file share access. This synchronization ensures that the identities are available in both environments, enabling seamless authentication.

2. Enable AD DS authentication

- To enable AD DS authentication over SMB for Azure file shares, you need to register your storage account with AD DS and then set the required domain properties on the storage account.

- You can think of this process as if it were like creating an account representing an on-premises Windows file server in your AD DS.

3. Assign share and directory permissions

- After enabling AD DS authentication, you must configure share-level permissions in order to get access to your file shares. The Azure RBAC share-level permissions act as a high-level gatekeeper that determines whether a user can access the share.

- With directory permissions, you can configure proper Windows ACLs at the root, directory, or file level, to take advantage of granular access control.

4. Mount file share with AD credentials

Hence, the correct sequence is:

1. Sync on-premises AD with Microsoft Entra Connect

2. Enable AD DS authentication

3. Assign share and directory permissions

4. Mount file share with AD credentials

References:

https://docs.microsoft.com/en-us/azure/storage/files/storage-files-introduction

https://docs.microsoft.com/en-us/azure/storage/files/storage-files-identity-auth-active-directory-enable

Check out this Azure Files Cheat Sheet:

https://tutorialsdojo.com/azure-file-storage/

Azure Blob vs Disk vs File Storage:

https://tutorialsdojo.com/azure-blob-vs-disk-vs-file-storage/

7 Your organization has a standard general-purpose v2 storage account with an access tier of Hot. The files uploaded to the storage account are infrequently accessed by your colleagues.

You were tasked with modifying the storage account with the following requirements:

- Inactive data must automatically transition to the archive tier after 120 days.
- Data uploaded must be accessed instantly, provided that it has not been transitioned to the archive tier yet.

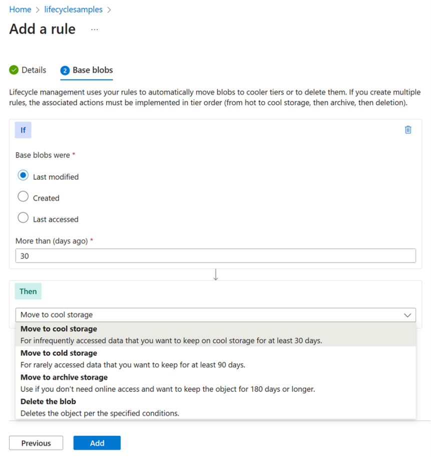
- Minimize costs.
- Minimize administrative effort.

Which actions should you perform? (Select TWO.)

(view) 1 0 1 00:00:00

☑ Create an Azure Function to move the inactive data to the archive tier after 120 days of inactivity.
☑ Create a lifecycle management rule to move the inactive data to the Archive tier after 120 days of inactivity.
☐ Manually copy the inactive data using the `Copy Blob` operation to the archive tier after 120 days of inactivity.
☐ Set the default access tier of the storage account to the Cool tier.
☐ Set the default access tier of the storage account to the Archive tier.
☐ Automatically archive data on upload.

An **Azure Storage Account** contains all of your Azure Storage data objects: blobs, files, queues, tables, and disks. The storage account provides a unique namespace for your Azure Storage data that is accessible from anywhere in the world over HTTP or HTTPS. Data in your Azure storage account is durable and highly available, secure, and massively scalable.



Home > lifecyclesamples >

Add a rule ···

✓ Details ② Base blobs

Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).

If

Base blobs were *

● Last modified
○ Created
○ Last accessed

More than (days ago) *

30

Then

Move to cool storage

Move to cool storage
For infrequently accessed data that you want to keep on cool storage for at least 30 days.
Move to cold storage
For rarely accessed data that you want to keep for at least 90 days.
Move to archive storage
Use if you don't need online access and want to keep the object for 180 days or longer.
Delete the blob
Deletes the object per the specified conditions.

Previous Add

Data sets have unique lifecycles. Early in the lifecycle, people access some data often. However, the need for access often drops drastically as the data ages. Some data remains idle in the cloud and is rarely accessed once stored.

Some data sets expire days or months after creation, while other data sets are actively read and modified throughout their lifetimes. Azure Storage **lifecycle management** offers a rule-based policy that you can use to transition blob data to the appropriate access tiers or to expire data at the end of the data lifecycle.

Storage accounts have a default access tier setting that indicates in which online tier a new blob is created. The default access tier setting can be either hot or cool only. The behavior of this setting is slightly different depending on the type of storage account:

Since the scenario states that your colleagues infrequently access the data, this means you do not need to store your data in the Hot tier. You can park the data in the Cool tier and automatically transition it to Archive using the data lifecycle.

Hence, the correct answers are:

- Create a lifecycle management rule to move the inactive data to the Archive tier after 120 days of inactivity.

- Set the default access tier of the storage account to the Cool tier.

The statement that says: **Create an Azure Function to move the inactive data to the archive tier after 120 days of inactivity** is incorrect because you can achieve the same goal using lifecycle management. Remember that one of the requirements is minimizing administrative effort.

The statement that says: **Set the default access tier of the storage account to the Archive tier** is incorrect because the supported default access tiers for storage accounts are Hot and Cool tiers. What you can do is move the data to the cool tier if your data is infrequently accessed and then create a lifecycle policy to transition unmodified data to the Archive tier after a set amount of time.

The statement that says: **Manually copy the inactive data using the `Copy Blob` operation to the archive tier after 120 days of inactivity** is incorrect because manually copying the inactive data to the Archive tier is a tedious task if you have thousands of data. One of the requirements states that you must lessen the administrative effort. Use lifecycle management instead.

The statement that says: **Automatically archive data on upload** is incorrect. One of the requirements states that you need your data that is not in the archive tier to be accessible instantly. Data in the Archive tier takes hours before you can access it.

References:

https://docs.microsoft.com/en-us/azure/storage/common/storage-account-overview

https://docs.microsoft.com/en-us/azure/storage/blobs/lifecycle-management-overview

https://learn.microsoft.com/en-us/azure/storage/blobs/lifecycle-management-policy-configure?tabs=azure-portal

Check out this Azure Storage Overview Cheat Sheet:

https://tutorialsdojo.com/azure-storage-overview/

8 You are configuring a blob container's access policy within an Azure storage account. You want to set multiple named access policies for fine-grained control and flexibility.

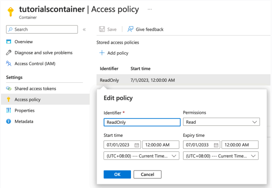
What is the maximum number of named access policies you can create for a blob container?

(view) 1 0 1 00:00:00

○ 1
○ 5
● 10
○ 20

A **stored access policy** provides an additional level of control over service-level shared access signatures (SASs) on the server side. Establishing a stored access policy serves to group shared access signatures and to provide additional restrictions for signatures that are bound by the policy.

With a container access policy, you can grant or revoke permissions for specific operations on blobs, such as read, write, delete, list, and more. The key benefit of using a container access policy is that it offers a more targeted and controlled approach to managing access to individual blobs within the container without the need to modify the storage account's shared access signature (SAS) settings.



Stored access policies

Identifier	Start time
ReadOnly	7/1/2023, 12:00:00 AM

Edit policy

Identifier *: ReadOnly
Permissions: Read
Start time: 07/01/2023 12:00:00 AM (UTC+08:00) --- Current Time...
Expiry time: 07/01/2033 12:00:00 AM (UTC+08:00) --- Current Time...

You can set a maximum of five access policies on a container, table, queue, or share at a time. Each SignedIdentifier field, with its unique Id field, corresponds to one access policy. Trying to set more than five access policies at one time causes the service to return status code 400 (Bad Request).

Hence, the correct answer is: **5.**

References:

https://learn.microsoft.com/en-us/azure/storage/common/storage-account-overview

https://learn.microsoft.com/en-us/rest/api/storageservices/define-stored-access-policy

Tutorials Dojo's AZ-305 Microsoft Azure Solutions Architect Expert Exam Study Guide:

https://tutorialsdojo.com/az-305-microsoft-azure-solutions-architect-expert-exam-study-guide/

Sub-Total:	13	5 (38.46%)	8 (61.54%)	00:00:00	38.46%

Category: **AZ-104 - Implement and Manage Virtual Networking**

5 You have an Azure virtual network named `TDVnet1` that contains the following subnets shown below:

Name	Connected virtual machines
TDSub1	TD1, TD2
TDSub2	TD3, TD4
TDSub3	TD5, TD6

You plan to create a network security group for your virtual machines.

Due to regulatory compliance, you must meet the following requirements:

1. Virtual machines in `TDSub2` and `TDSub3` must have HTTPS traffic from the Internet.
2. Remote Desktop connections from the public Internet must only access `TD1` .
3. All traffic between `TD1` and `TD2` must be allowed.
4. Restrict all other external network traffic from accessing `TDVnet1` .

What is the minimum number of network security groups that you should provision to satisfy the requirements above?

(view) 1 0 1 00:00:00

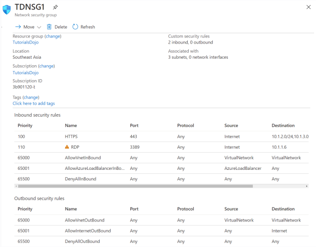
○ 1

○ 3

○ 5

○ 6

Azure Network Security Group is used to filter network traffic to and from Azure resources in an Azure virtual network. A network security group contains security rules that allow or deny inbound network traffic to, or outbound network traffic from, several types of Azure resources. For each rule, you can specify source and destination, port, and protocol.



Resource group (change): TutorialsDojo
Location: Southeast Asia
Subscription (change): TutorialsDojo
Subscription ID: 3b901120-ł
Tags (change)
Custom security rules: 2 inbound, 0 outbound
Associated with: 3 subnets, 0 network interfaces

Inbound security rules

Priority	Name	Port	Protocol	Source	Destination
100	HTTPS	443	Any	Internet	10.1.2.0/24,10.1.3.0
110	⚠ RDP	3389	Any	Internet	10.1.1.6
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowAzureLoadBalancerInBo...	Any	Any	AzureLoadBalancer	Any
65500	DenyAllInBound	Any	Any	Any	Any

Outbound security rules

Priority	Name	Port	Protocol	Source	Destination
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowInternetOutBound	Any	Any	Any	Internet
65500	DenyAllOutBound	Any	Any	Any	Any

Network Security Groups can be attached to multiple subnets and/or network interfaces. Unless you have a specific reason to, it is recommended that you associate a network security group to a subnet or a network interface, but not both.

In the image above, the requirements of the scenario are fully satisfied. You only need to create one network security group with multiple rules and associate it with `TDSub1` , `TDSub2` , and `TDSub3` .

1. Virtual machines in `TDSub2` and `TDSub3` must have HTTPS traffic from the Internet.
   - You can whitelist the address spaces of `TDSub2` and `TDSub3` in the **destination IP addresses/CIDR ranges** of an inbound security rule. This will force HTTPS traffic to only those subnets without allowing HTTPS traffic to `TDSub1` . See *priority 100* in the image above.
2. Remote Desktop connections from the Internet must access `TD1` .
   - Since there are two virtual machines in `TDSub1` and the requirement states that only `TD1` must have Remote Desktop connection, you cannot whitelist the address space of `TDSub1` in the destination IP addresses.
   - An alternative to this is whitelisting the IP address of `TD1` to the **destination IP addresses** when you create an inbound security rule. See *priority 110* in the image above.
3. All traffic between `TD1` and `TD2` must be allowed.
   - When you create a network security group, the default rules of a network security group always allow traffic coming from WITHIN the virtual network. No action is needed from your side.
4. Restrict all other external network traffic from accessing `TDVnet1.`
   - The default rules of a network security group explicitly deny all incoming traffic. No action is needed from your side.

Hence, the correct answer is: **1.**

3, 5, and 6 are incorrect because you only need to create one network security group with multiple rules to satisfy the requirements of the scenario.

References:

https://docs.microsoft.com/en-us/azure/virtual-network/virtual-networks-overview

https://learn.microsoft.com/en-us/azure/virtual-network/network-security-group-how-it-works

Check out this Azure Virtual Network Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-network-vnet/

6 You have an Azure subscription that contains an Azure DNS zone named `tutorialsdojo.com` . The company also uses DirectAccess to provide remote connectivity for its on-premises users.

Your Azure DNS zone contains several record sets, including an SOA record and an A record.

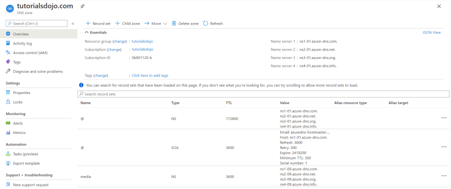
There is a requirement to delegate a subdomain named `portal.tutorialsdojo.com` to another Azure DNS zone.

What solution would satisfy the requirement?

(view) 1 0 1 00:00:00

- ● Navigate to `tutorialsdojo.com` and add a PTR record named `portal` .
- ○ Navigate to `tutorialsdojo.com` and add an NS record named `portal` .
- ○ Navigate to `tutorialsdojo.com` and add a CNAME record named `portal` .
- ○ Navigate to `tutorialsdojo.com` and add a TXT record named `portal` .

Azure DNS is a hosting service for DNS domains that provides name resolution by using Microsoft Azure infrastructure. By hosting your domains in Azure, you can manage your DNS records by using the same credentials, APIs, tools, and billing as your other Azure services.



Resource group (change) : tutorialsdojo

Subscription (change) : tutorialsdojo

Subscription ID : 3b901120-b

Tags (change) : Click here to add tags

Name server 1 : ns1-01.azure-dns.com.

Name server 2 : ns2-01.azure-dns.net.

Name server 3 : ns3-01.azure-dns.org.

Name server 4 : ns4-01.azure-dns.info.

You can search for record sets that have been loaded on this page. If you don't see what you're looking for, you can try scrolling to allow more record sets to load.

Name	Type	TTL	Value	Alias resource type	Alias target
@	NS	172800	ns1-01.azure-dns.com. ns2-01.azure-dns.net. ns3-01.azure-dns.org. ns4-01.azure-dns.info.		
@	SOA	3600	Email: azuredns-hostmaster.... Host: ns1-01.azure-dns.com. Refresh: 3600 Retry: 300 Expire: 2419200 Minimum TTL: 300 Serial number: 1		
media	NS	3600	ns1-09.azure-dns.com. ns2-09.azure-dns.net. ns3-09.azure-dns.org. ns4-09.azure-dns.info.		

You can use the Azure portal to delegate a DNS subdomain. For example, if you own the `tutorialsdojo.com` domain, you can delegate a subdomain called **portal** to another, separate zone that you can administer separately from the `tutorialsdojo.com` zone.

To delegate an Azure DNS subdomain, you must first delegate your public domain to Azure DNS. Once your domain is delegated to your Azure DNS zone, you can delegate your subdomain.

You can delegate a subdomain by doing the following:

1. Create a new Azure DNS zone named `portal.tutorialsdojo.com` . Copy down the four nameservers as you will need them for step 2.
2. Navigate to the `tutorialsdojo.com` DNS zone and add an NS record named **portal**. Under records, enter the four nameservers from `portal.tutorialsdojo.com` and click ok.
3. To verify your work, open a PowerShell window and type `nslookup portal.tutorialsdojo.com`

Hence, the correct answer is: **Navigate to `tutorialsdojo.com` and add an NS record named `portal` .**

The following statements are incorrect because PTR, CNAME, and TXT records are not used to delegate an Azure DNS subdomain.

- Navigate to `tutorialsdojo.com` and add a PTR record named `portal` .

- Navigate to `tutorialsdojo.com` and add a CNAME record named `portal` .

- Navigate to `tutorialsdojo.com` and add a TXT record named `portal.`

References:

https://docs.microsoft.com/en-us/azure/dns/dns-overview

https://docs.microsoft.com/en-us/azure/dns/delegate-subdomain

Check out this Azure DNS Cheat Sheet:

https://tutorialsdojo.com/azure-dns/

8 Your company is currently hosting a mission-critical application in an Azure virtual machine that resides in a virtual network named `TDVnet1` . You plan to use Azure ExpressRoute to allow the web applications to connect to the on-premises network, and you are using Network Performance Monitor to monitor the latency between Azure and the on-premises environment.

Due to compliance requirements, you need to ensure that in the event your ExpressRoute fails, the connectivity between `TDVnet1` and your on-premises network will remain available.

The solution must utilize a site-to-site VPN between `TDVnet1` and the on-premises network. The solution should also be cost-effective.

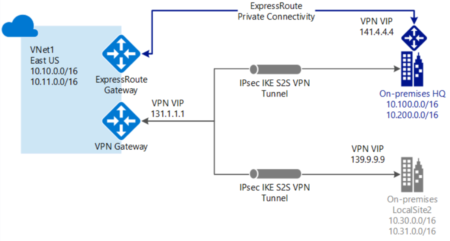
Which actions should you implement? (Select THREE.)

Each correct answer presents part of the solution.

(view) 1 0 1 00:00:00

- ☐ Configure an Azure Virtual WAN.
- ☐ Configure a VPN gateway with `VpnGw1` as its SKU.
- ☑ Configure a VPN gateway with `Basic` as its SKU.
- ☑ Configure a local network gateway.
- ☑ Configure a connection.
- ☐ Configure a Virtual Hub with a Standard SKU.

A **VPN gateway** is a specific type of virtual network gateway that is used to send encrypted traffic between an Azure virtual network and an on-premises location over the public Internet. You can also use a VPN gateway to send encrypted traffic between Azure virtual networks over the Microsoft network. Each virtual network can have only one VPN gateway. However, you can create multiple connections to the same VPN gateway. When you create multiple connections to the same VPN gateway, all VPN tunnels share the available gateway bandwidth.



A site-to-site VPN gateway connection is used to connect your on-premises network to an Azure virtual network over an IPsec/IKE (IKEv1 or IKEv2) VPN tunnel. This type of connection requires a VPN device located on-premises that has an externally facing public IP address assigned to it.

Configuring Site-to-Site VPN and ExpressRoute coexisting connections has several advantages:

- You can configure a Site-to-Site VPN as a secure failover path for ExpressRoute.
- Alternatively, you can use Site-to-Site VPNs to connect to sites that are not connected through ExpressRoute.

To create a site-to-site connection, you need to do the following:

- Provision a virtual network
- Provision a VPN gateway
- Provision a local network gateway
- Provision a VPN connection
- Verify the connection
- Connect to a virtual machine

Take note that since you have already deployed an ExpressRoute, you do not need to create a virtual network and gateway subnet as these are prerequisites in creating an ExpressRoute.

Hence, the correct answers are:

- **Configure a VPN gateway with `VpnGw1` as its SKU.**
- **Configure a local network gateway.**
- **Configure a connection.**

The option that says: **Configure an Azure Virtual WAN** is incorrect. Azure Virtual WAN is a service for creating a global-wide network, which is more suitable for complex network topologies. In the scenario, this architecture is unnecessary because the setup only requires a site-to-site VPN as a backup for ExpressRoute and does not demand a Virtual WAN architecture.

The option that says: **Configure a VPN gateway with `Basic` as its SKU** is incorrect. Although one of the requirements is to minimize costs, the coexisting connection for ExpressRoute and site-to-site VPN connection does not primarily support a `Basic` SKU. The bare minimum for a coexisting connection is `VpnGw1` .

The option that says: **Configure a Virtual Hub with a Standard SKU** is incorrect because it is part of the Azure Virtual WAN architecture, which is not needed here. Using a Virtual Hub would increase cost and complexity without adding value for this specific requirement.

References:

https://docs.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-howto-site-to-site-resource-manager-portal

https://docs.microsoft.com/en-us/azure/expressroute/expressroute-howto-coexist-resource-manager

https://docs.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-about-vpngateways#gwsku

Check out this Azure VPN Gateway Cheat Sheet:

https://tutorialsdojo.com/azure-vpn-gateway/

Sub-Total:	15	12 (80%)	3 (20%)	00:00:00	80%

Category: **AZ-104 - Manage Azure Identities and Governance**

2	You purchased a Microsoft Entra ID Premium P2 license. The company already uses Azure Arc-enabled servers to manage its hybrid machine inventory across on-premises and cloud. You plan to add a local administrator to manage all the computers and devices that will join your domain. What do you need to configure in Microsoft Entra to satisfy this requirement? (view)	1	0	1	00:00:00

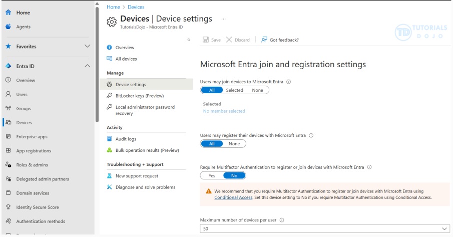
○ Manage device settings through the Devices blade.

○ Configure OAuth 2.0 authorization endpoint.

○ Enable app registrations in user settings.

○ Configure group naming policy.

Microsoft Entra ID is Microsoft's cloud-based identity and access management service, which helps your employees sign in and access resources from external resources such as Microsoft 365, the Azure portal, and thousands of other SaaS applications. It can also be used on internal resources, such as apps on your corporate network and intranet, along with any cloud apps developed by your own organization.



Devices | Device settings

Microsoft Entra join and registration settings

Users may join devices to Microsoft Entra: [✓ All] Selected None

Selected: No member selected

Users may register their devices with Microsoft Entra: [✓ All] None

Require Multifactor Authentication to register or join devices with Microsoft Entra: Yes [✓ No]

We recommend that you require Multifactor Authentication to register or join devices with Microsoft Entra using Conditional Access. Set this device setting to No if you require Multifactor Authentication using Conditional Access.

Maximum number of devices per user: 50

To add local administrators who will manage joined devices in Microsoft Entra, you must configure the settings shown in the image above. You can select the users who are granted local administrator rights on a device. These users will be added to the Device Administrators' role in Microsoft Entra. By default, global administrators in Microsoft Entra and device owners are granted local administrator rights. This configuration is a premium edition capability, and it is available through products such as Microsoft Entra Premium or the Enterprise Mobility Suite (EMS).

Hence, the correct answer is: **Manage device settings through the Devices blade.**

The option that says: **Configure OAuth 2.0 authorization endpoint** is incorrect because this relates to how applications authenticate and authorize access to Microsoft Entra ID resources, typically in scenarios involving delegated permissions or token issuance. It has no relevance to assigning local administrator rights on domain-joined devices, which is a device-level configuration, not an identity protocol concern.

The option that says: **Enable app registrations in user settings** is incorrect. If this option is enabled, non-admin users can register custom-developed applications within the directory. This option is not needed. Take note that the requirement in the scenario is to add local administrators and not to register applications.

The option that says: **Configure group naming policy** is incorrect. This option only allows you to add a specific prefix or suffix to the group name and alias of any Microsoft 365 group created by users. Group naming policy is not needed in the scenario because you only need to add local administrators to your domain.

References:

https://learn.microsoft.com/en-us/entra/identity/devices/manage-device-identities

https://learn.microsoft.com/en-us/entra/identity/devices/assign-local-admin#manage-the-device-administrator-role

Check out this Microsoft Entra ID Cheat Sheet:

https://tutorialsdojo.com/microsoft-entra-id/

5 Note: This item is part of a series of case study questions with the exact same scenario but has a different technical requirement. Each one in the series has a unique solution that may or may not comply with the requirements specified in the scenario.

Overview	Adatum Corporation is an insurance company that has a total of 5,000 employees with its headquarters located in Singapore and three satellite offices in Tokyo, Seattle, and London.
Existing Environment	
Planned Changes	
Technical Requirements	

Question

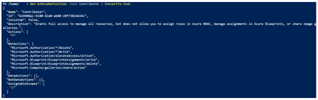
Which command should you run?

(view) 1 0 1 00:00:00

- ○ `Get-AzRoleAssignment -Name Contributor | ConvertTo-Json`
- ○ `Get-AzRoleAssignment -Name Contributor | ConvertFrom-Json`
- ● `Get-AzRoleDefinition -Name Contributor | ConvertTo-Json`
- ○ `Get-AzRoleDefinition -Name Contributor | ConvertFrom-Json`

Access management for cloud resources is a critical function for any organization that is using the cloud. Azure **role-based access control** (Azure RBAC) helps you manage who has access to Azure resources, what they can do with those resources, and what areas they have access to.

Azure RBAC is an authorization system built on Azure Resource Manager that provides fine-grained access management of Azure resources.



```
PS /home/    > Get-AzRoleDefinition -Name Contributor | ConvertTo-Json
{
  "Name": "Contributor",
  "Id": "b24988ac-6180-42a0-ab88-20f7382dd24c",
  "IsCustom": false,
  "Description": "Grants full access to manage all resources, but does not allow you to assign roles in Azure RBAC, manage assignments in Azure Blueprints, or share image g
alleries.",
  "Actions": [
    "*"
  ],
  "NotActions": [
    "Microsoft.Authorization/*/Delete",
    "Microsoft.Authorization/*/Write",
    "Microsoft.Authorization/elevateAccess/Action",
    "Microsoft.Blueprint/blueprintAssignments/write",
    "Microsoft.Blueprint/blueprintAssignments/delete",
    "Microsoft.Compute/galleries/share/action"
  ],
  "DataActions": [],
  "NotDataActions": [],
  "AssignableScopes": [
    "/"
  ]
}
```

If the Azure built-in roles don't meet the specific needs of your organization, you can create your own custom roles. Just like built-in roles, you can assign custom roles to users, groups, and service principals at management group, subscription, and resource group scopes.

Take note that in this scenario, you need to create a custom role named `AdatumAdministrator` that is based on the built-in policy Contributor role. You need to retrieve the JSON format file of the Contributor role so that you can customize it to your needs.

To retrieve the JSON string of the Contributor role, you need to use the command:

- `Get-AzRoleDefinition -Name <role_name> | ConvertTo-Json`

Hence, the correct answer is: **`Get-AzRoleDefinition -Name Contributor | ConvertTo-Json`**

`Get-AzRoleDefinition -Name Contributor | ConvertFrom-Json` is incorrect because the `ConvertFrom-Json` cmdlet just converts your JSON string to a PSCustomObject object that has a property for each field in the JSON string. Take note that you need to retrieve the JSON role so that you can customize it to your needs.

The following options are incorrect because the `Get-AzRoleAssignment` simply allows you to list Azure RBAC role assignments at the specified scope. By default, it lists all role assignments in the selected Azure subscription. You have to use the respective parameters to list assignments to a specific user, or to list assignments on a specific resource group or resource.

- **`Get-AzRoleAssignment -Name Contributor | ConvertTo-Json`**
- **`Get-AzRoleAssignment -Name Contributor | ConvertFrom-Json`**

References:

https://docs.microsoft.com/en-us/azure/role-based-access-control/overview

https://docs.microsoft.com/en-us/azure/role-based-access-control/custom-roles-powershell

Check out this Azure Role-Based Access Control (RBAC) Cheat Sheet:

https://tutorialsdojo.com/azure-role-based-access-control-rbac

6 Note: This item is part of a series of questions with the exact same scenario but with a different proposed answer. Each one in the series has a unique solution that may, or may not, comply with the requirements specified in the scenario.

You created a new Microsoft Entra group for Network Administrators in your organization Azure Subscription.

You need to make sure that the users in the group can enable Traffic Analytics and visualize traffic distribution.

Solution: Assign a Contributor role to the group.

Does the solution meet the goal?

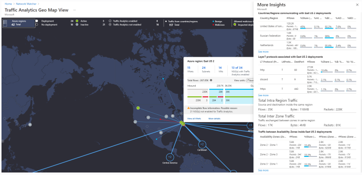
(view) 1 0 1 00:00:00

- ● Yes
- ○ No

Traffic analytics is a cloud-based solution that provides visibility into user and application activity in your cloud networks. Specifically, traffic analytics analyzes Azure Network Watcher network security group (NSG) flow logs to provide insights into traffic flow in your Azure cloud.

With traffic analytics, you can:

- Visualize network activity across your Azure subscriptions.
- Identify hot spots.
- Secure your network by using information about the following components to identify threats: Open ports, Applications that attempt to access the internet, and VMs that connect to rogue networks.
- Optimize your network deployment for performance and capacity by understanding traffic flow patterns across Azure regions and the internet.
- Pinpoint network misconfigurations that can lead to failed connections in your network.



Total Intra Region Traffic

Source and destination inside the same region

Flows : 35K Bytes : 116MB Packets : 228K

Total Inter Zone Traffic

Traffic exchanged between zones in same region

Flows : 17K Bytes : 4MB Packets : 81K

To enable traffic analytics, your account must have any of the following Azure roles at the subscription scope: owner, contributor, or network contributor.

But before you use traffic analytics, ensure your environment meets the following requirements:

- A Network Watcher enabled subscription.
- Network Security Group (NSG) flow logs enabled for the NSGs you want to monitor.
- An Azure Storage account to store raw flow logs.
- An Azure Log Analytics workspace with read and write access.

Going back to the given solution, the users in the group can visualize the traffic distribution by assigning a Contributor role to the group. A Contributor role can manage all resources but does not allow you to assign roles in Azure RBAC, manage assignments in Azure Blueprints, or share image galleries.

Hence, the correct answer is: **Yes.**

References:

https://learn.microsoft.com/en-us/azure/network-watcher/traffic-analytics-faq#what-are-the-prerequisites-to-use-traffic-analytics-

https://learn.microsoft.com/en-us/azure/network-watcher/traffic-analytics

Check out this Microsoft Entra ID Cheat Sheet:

https://tutorialsdojo.com/microsoft-entra-id/

8 *Note: This item is part of a series of questions with the exact same scenario but with a different proposed answer. Each one in the series has a unique solution that may, or may not, comply with the requirements specified in the scenario.*

Your organization has a Microsoft Entra ID subscription that is associated with the directory TD-Siargao.

You have been tasked to implement a conditional access policy.

The policy must require the DevOps group to use multi-factor authentication and a hybrid Microsoft Entra joined device when connecting to Microsoft Entra from untrusted locations.

Solution: Create a conditional access policy and enforce grant control.

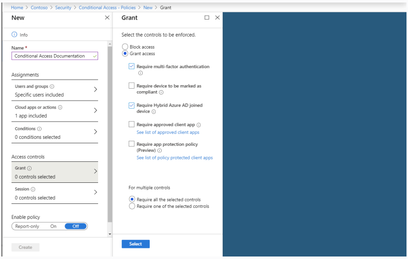
Does the solution meet the goal?

(view) 1 0 1 00:00:00

○ Yes

○ No

Microsoft Entra ID enterprise identity service provides single sign-on and multi-factor authentication to help protect your users from 99.9 percent of cybersecurity attacks. The single sign-on is an authentication method that simplifies access to your apps from anywhere. While conditional access and multi-factor authentication help protect and govern access to your resources.


Home > Contoso > Security > Conditional Access - Policies > New > Grant
New
Info
Assignments
Access controls
Create
Grant
Select

Name * Conditional Access Documentation

Users and groups: Specific users included

Cloud apps or actions: 1 app included

Conditions: 0 conditions selected

Grant: 0 controls selected

Session: 0 controls selected

Enable policy: Report-only On Off

Select the controls to be enforced.

○ Block access

● Grant access

☑ Require multi-factor authentication

☐ Require device to be marked as compliant

☑ Require Hybrid Azure AD joined device

☐ Require approved client app
See list of approved client apps

☐ Require app protection policy (Preview)
See list of policy protected client apps

For multiple controls

● Require all the selected controls

○ Require one of the selected controls

With conditional access, you can implement automated access-control decisions for accessing your cloud apps based on conditions. Conditional access policies are enforced after the first-factor authentication has been completed. It's not intended to be a first-line defense against denial-of-service (DoS) attacks, but it uses signals from these events to determine access.

There are two types of access controls in a conditional access policy:

1. Grant - enforces grant or block access to resources.
2. Session - enable limited experiences within specific cloud applications

Going back to the scenario, the requirement is to enforce a policy to the members of the DevOps group to use MFA and a hybrid Microsoft Entra joined device when connecting to Microsoft Entra from untrusted locations. The given solution is to enforce grant access control. If you check the image above, the grant control satisfies this requirement.

Hence, the correct answer is: **Yes.**

References:

https://learn.microsoft.com/en-us/entra/identity/conditional-access/overview

https://learn.microsoft.com/en-us/entra/identity/conditional-access/howto-conditional-access-policy-all-users-mfa

https://learn.microsoft.com/en-us/entra/identity/conditional-access/concept-conditional-access-grant

Check out this Microsoft Entra ID Cheat Sheet:

https://tutorialsdojo.com/microsoft-entra-id/

9 *Note: This item is part of a series of questions with the exact same scenario but with a different proposed answer. Each one in the series has a unique solution that may, or may not, comply with the requirements specified in the scenario.*

Your organization has an Azure AD subscription that is associated with the directory TD-Siargao.

You have been tasked to implement a conditional access policy.

The policy must require the DevOps group to use multi-factor authentication and a hybrid Azure AD joined device when connecting to Azure AD from untrusted locations.

Solution: Create a conditional access policy and enforce session control.

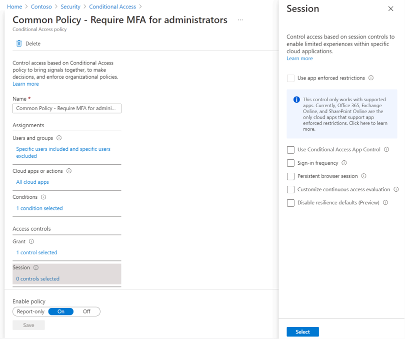
Does the solution meet the goal?

(view) 1 0 1 00:00:00

○ Yes

● No

Azure Active Directory (Azure AD) enterprise identity service provides single sign-on and multi-factor authentication to help protect your users from 99.9 percent of cybersecurity attacks. The single sign-on is an authentication method that simplifies access to your apps from anywhere. While conditional access and multi-factor authentication help protect and govern access to your resources.



Common Policy - Require MFA for administrators

Conditional Access policy

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies. Learn more

Name *: Common Policy - Require MFA for admini...

Users and groups: Specific users included and specific users excluded

Cloud apps or actions: All cloud apps

Conditions: 1 condition selected

Grant: 1 control selected

Session: 0 controls selected

Enable policy: Report-only | On | Off

Session

Control access based on session controls to enable limited experiences within specific cloud applications. Learn more

☐ Use app enforced restrictions

This control only works with supported apps. Currently, Office 365, Exchange Online, and SharePoint Online are the only cloud apps that support app enforced restrictions. Click here to learn more.

☐ Use Conditional Access App Control

☐ Sign-in frequency

☐ Persistent browser session

☐ Customize continuous access evaluation

☐ Disable resilience defaults (Preview)

With conditional access, you can implement automated access-control decisions for accessing your cloud apps based on conditions. Conditional access policies are enforced after the first-factor authentication has been completed. It's not intended to be a first-line defense against denial-of-service (DoS) attacks, but it uses signals from these events to determine access.

There are two types of access controls in a conditional access policy:

1. Grant - enforces grant or block access to resources.
2. Session - enable limited experiences within specific cloud applications

Going back to the scenario, the requirement is to enforce a policy to the members of the DevOps group to use MFA and a hybrid Azure AD joined device when connecting to Azure AD from untrusted locations. The given solution is to enforce session access control. If you check the image above, the session control doesn't have options to require the use of MFA and AD joined devices.

Hence, the correct answer is: **No.**

References:

https://docs.microsoft.com/en-us/azure/active-directory/conditional-access/overview

https://docs.microsoft.com/en-us/azure/active-directory/conditional-access/howto-conditional-access-policy-all-users-mfa

https://docs.microsoft.com/en-us/azure/active-directory/conditional-access/concept-conditional-access-grant

Check out this Azure Active Directory Cheat Sheet:

https://tutorialsdojo.com/azure-active-directory-azure-ad/

10 You are managing an Azure subscription that includes 150 virtual machines. These virtual machines are generated and terminated frequently as part of your operations. To optimize storage costs, you need to identify unattached disks that can be removed.

What action should you take?

(view) 1 0 1 00:00:00

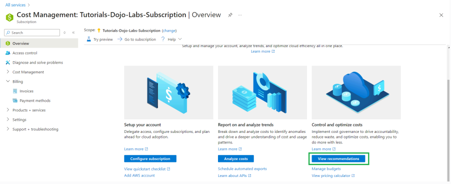
○ Check the Cost Analysis from Azure Cost Management.

○ Enable diagnostic settings in Azure Monitor.

○ Explore the Account Management properties in Microsoft Azure Storage Explorer.

● Go to Advisor Recommendations in Azure Cost Management.

Azure Advisor provides customized best practices and suggestions for optimizing your Azure resources. It offers detailed advice on costs, security, dependability, operational excellence, and performance.



Cost Management: Tutorials-Dojo-Labs-Subscription | Overview

Scope: Tutorials-Dojo-Labs-Subscription (change)

Setup and manage your account, analyze trends, and optimize cloud efficiency all in one place. Learn more

Setup your account

Delegate access, configure subscriptions, and plan ahead for cloud adoption.

Report on and analyze trends

Break down and analyze costs to identify anomalies and drive a deeper understanding of cost and usage patterns.

Control and optimize costs

Implement cost governance to drive accountability, reduce waste, and optimize costs, enabling you to do more with less.

To identify unattached disks, Azure Advisor analyzes your environment and provides recommendations for disks that are not currently associated with any virtual machines (VMs). These unattached disks can be safely deleted to reduce storage costs. Viewing Advisor Recommendations within Azure Cost Management will give you insights into which disks are unattached and can be removed, helping you optimize your storage resources efficiently.

Hence, the correct answer is: **Go to Advisor Recommendations in Azure Cost Management.**

The option that says: **Check the Cost Analysis from Azure Cost Management** is incorrect because Cost Analysis only provides insights into overall spending and cost trends but does not specifically identify unattached disks. It is designed to help you understand where your money is going, not to identify unused or unnecessary resources.

The option that says: **Enable diagnostic settings in Azure Monitor** is incorrect. While diagnostic settings in Azure Monitor can provide valuable monitoring data, they do not specifically identify unattached disks. Diagnostic settings are aimed at performance and health monitoring rather than cost optimization for unattached disks.

The option that says: **Explore the Account Management properties in Microsoft Azure Storage Explorer** is incorrect because this tool is primarily used for managing storage accounts and exploring data stored in Azure. Although you can view and manage various properties of your storage accounts, it does not provide specific recommendations or insights on unattached disks. This tool is more suited for managing storage objects and data than identifying optimization opportunities.

References:

https://learn.microsoft.com/en-us/azure/advisor/advisor-cost-recommendations

https://learn.microsoft.com/en-us/azure/cost-management-billing/cost-management-billing-overview

Check out this Azure Advisor Cheat Sheet:

https://tutorialsdojo.com/azure-advisor/

Sub-Total:	10	4 (40%)	6 (60%)	00:00:00	**40%**

Category: **AZ-104 - Monitor and Maintain Azure Resources**

1 You are managing an Azure subscription hosting a virtual machine named <code>td-cebu-vm1</code>, which supports critical workloads.

The IT team has reported network performance issues affecting this VM. You must enable packet capture and conduct detailed network diagnostics to troubleshoot and resolve the problem.

To complement your investigation, you also plan to use Azure Traffic Analytics to gain insights into traffic patterns and detect anomalies.

Which action should you perform first to enable packet capture and begin diagnostics?

(view) 1 0 1 00:00:03

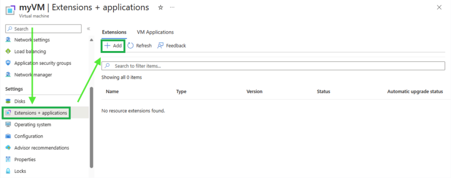
○ Install the Azure Performance Diagnostics Agent on `td-cebu-vm1` .

○ Install the Azure Network Watcher Agent VM extension on `td-cebu-vm1` .

○ Enable Microsoft Defender for Servers.

○ Install Azure Monitor agent extension on `td-cebu-vm1` .

The **Azure Network Watcher Agent VM extension** is a component that can be installed on Azure virtual machines to enable advanced network monitoring and diagnostic capabilities. This extension works with Azure Network Watcher to provide features such as packet capture, IP flow verification, and detailed network performance analysis. It acts as a bridge between the virtual machine and the Network Watcher service, allowing for deep insights into network traffic and behavior.



Showing all 0 items

No resource extensions found.

Installing the Network Watcher Agent VM extension on `td-cebu-vm1` would provide the necessary tools to address the reported network performance issues. This extension enables packet capture functionality, which is crucial for diagnosing complex network problems. By capturing and analyzing network packets, IT teams can identify bottlenecks, latency issues, or unexpected traffic patterns affecting the VM's performance. Additionally, the extension facilitates other network diagnostic features that can help pinpoint the root cause of the performance problems.

Therefore, the correct answer is: **Install the Azure Network Watcher Agent VM extension on `td-cebu-vm1`** .

The option that says: **Install the Azure Performance Diagnostics Agent on `td-cebu-vm1`** is incorrect because Azure Performance Diagnostics Agent is primarily designed to collect performance-related logs and metrics for troubleshooting boot issues or application performance problems on Azure VMs. However, it does not provide network diagnostics features like packet capture or advanced network performance troubleshooting.

The option that says: **Enable Microsoft Defender for Servers** is incorrect because Microsoft Defender for Servers is primarily a security solution that provides threat detection and advanced defenses. It doesn't offer the specific network diagnostics and packet capture capabilities required.

The option that says: **Install Azure Monitor agent extension on `td-cebu-vm1`** is incorrect because the Azure Monitor agent is typically used for collecting monitoring data from the operating system and workloads on Azure and hybrid virtual machines. While it can collect network-related metrics, it doesn't provide the detailed packet capture functionality needed for this scenario.

References:

https://learn.microsoft.com/en-us/azure/network-watcher/network-watcher-overview

https://learn.microsoft.com/en-us/azure/network-watcher/network-watcher-agent-windows?tabs=portal

https://learn.microsoft.com/en-us/azure/network-watcher/network-watcher-agent-linux?tabs=portal

Check out these Azure Cheat Sheets:

https://tutorialsdojo.com/microsoft-azure-cheat-sheets/

2 Your company has an Azure subscription that contains the following resources:

Virtual Machine	Operating System	Location
TD1	Windows Server 2019	Southeast Asia
TD2	Windows Server 2016	Southeast Asia
TD3	Windows Server 2012	Southeast Asia

You have an Azure Recovery Services vault named `TDBackup1` that backs up `TD1` , `TD2` , and `TD3` daily without an Azure Backup Agent.

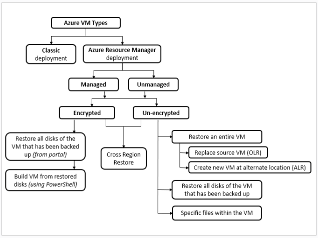
Select the correct answer from the drop-down list of options. Each correct selection is worth one point.

(view) 2 1 1 00:00:00

You can execute a file recovery operation TD2 to: **TD1, TD2, and TD3** (TD2 only)
Choices: TD1 only, TD2 only, TD3 only, TD1, TD2, and TD3

You can restore TD3 to: **TD3 only**
Choices: TD1 only, TD2 only, TD3 only, TD1, TD2, and TD3.

Azure Backup provides independent and isolated backups to guard against unintended destruction of the data on your VMs. Backups are stored in a Recovery Services vault with built-in management of recovery points. Configuration and scaling are simple, backups are optimized, and you can easily restore as needed.


Azure VM Types
Classic deployment
Azure Resource Manager deployment
Managed
Unmanaged
Encrypted
Un-encrypted
Restore all disks of the VM that has been backed up (from portal)
Build VM from restored disks (using PowerShell)
Cross Region Restore
Restore an entire VM
Replace source VM (OLR)
Create new VM at alternate location (ALR)
Restore all disks of the VM that has been backed up
Specific files within the VM

To **recover a specific file**, you must specify the recovery point of your backup and download a script that will mount the disks from the selected recovery point. After the script is successfully downloaded, make sure you have the right machine to execute this script.

When recovering files, you can't restore files to a previous or future operating system version. For example, you can't restore a file from a Windows Server 2016 VM to Windows Server 2012 or a Windows 8 computer. You can restore files from a VM to the same server operating system, or to the compatible client operating system.

You can restore a virtual machine with the following options:

- Create a new VM
- Restore Disk
- Replace existing disk (OLR)

As one of the restore options, you can **replace an existing VM disk** with the selected restore point. The current VM must exist. If it's been deleted, this option can't be used. Azure Backup takes a snapshot of the existing VM before replacing the disk, and stores it in the staging location you specify.

Existing disks connected to the VM are replaced with the selected restore point. The snapshot is copied to the vault and retained in accordance with the retention policy.

After the *Replace Disk* operation, the original disk is retained in the resource group. You can choose to manually delete the original disks if they aren't needed.

Therefore, you can perform file recovery to **TD2 only** because the operating systems of `TD1` and `TD3` are not compatible with `TD2`. You need to ensure that the machine you are recovering the file to meets the requirements before executing the script.

Conversely, you can restore `TD3` to **TD3 only** because you can not restore the disk of `TD3` to `TD1` and `TD2`. You can only restore a virtual machine by creating a new VM, restoring a disk, or replace the existing VM disk.

References:

https://docs.microsoft.com/en-us/azure/backup/backup-overview

https://docs.microsoft.com/en-us/azure/backup/backup-azure-arm-restore-vms

https://docs.microsoft.com/en-us/azure/backup/backup-azure-restore-files-from-vm

Check out this Azure Virtual Machines Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-machines/

4 Your company has an Azure subscription that has the following resources deployed:

Name	Service	Location
TDBackup1	Recovery Services Vault	Southeast Asia
TDAccount2	Storage account	Southeast Asia
TDAnalytics1	Log Analytics workspace	East Asia
TDAnalytics2	Log Analytics workspace	Southeast Asia
TDAnalytics3	Log Analytics workspace	Australia Central

There is a requirement that requires you to configure Azure Backup reports using `TDBackup1` to determine which backup items consume the most storage.

Which Log Analytics workspace can you use to store the backup reporting data?

(view) 1 0 1 00:00:00

6 Note: This item is part of a series of case study questions with the exact same scenario but has a different technical requirement. Each one in the series has a unique solution that may or may not comply with the requirements specified in the scenario.

Overview
Existing Environment
Planned Changes
Technical Requirements

Tutorials Dojo is an online learning portal for technology-related topics that empowers its users to upgrade their skills and career. Tutorials Dojo has users worldwide, ranging from the United States, Europe, and Asia.

Question

What should you create first?

(view) 1 0 1 00:00:00

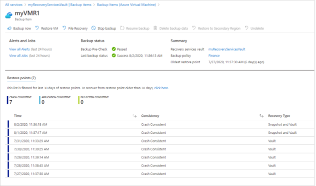
○ Backup policy

● Recovery Services Vault

○ Microsoft Azure Backup Server (MABS)

○ Recovery Plan

Azure Backup is a cost-effective, secure, one-click backup solution that's scalable based on your backup storage needs. The centralized management interface makes it easy to define backup policies and protect a wide range of enterprise workloads, including Azure Virtual Machines, SQL and SAP databases, and Azure file shares.


All services > myRecoveryServicesVault | Backup items > Backup Items (Azure Virtual Machine) >
myVMR1
Backup Item
Backup now
Restore VM
File Recovery
Stop backup
Resume backup
Delete backup data
Restore to Secondary Region
Undelete

Alerts and Jobs

View all Alerts (last 24 hours)

View all Jobs (last 24 hours)

Backup status

Backup Pre-Check ✓ Passed

Last backup status ✓ Success 8/2/2020, 11:36:13 AM

Summary

Recovery services vault myRecoveryServicesVault

Backup policy Finance

Oldest restore point 7/27/2020, 11:37:30 AM (6 day(s) ago)

Restore points (7)

This list is filtered for last 30 days of restore points. To recover from restore point older than 30 days, click here.

CRASH CONSISTENT 7

APPLICATION CONSISTENT 0

FILE-SYSTEM CONSISTENT 0

Time	Consistency	Recovery Type
8/2/2020, 11:36:18 AM	Crash Consistent	Snapshot and Vault
8/1/2020, 11:37:17 AM	Crash Consistent	Snapshot and Vault
7/31/2020, 11:33:29 AM	Crash Consistent	Vault
7/30/2020, 11:39:25 AM	Crash Consistent	Vault
7/29/2020, 11:39:14 AM	Crash Consistent	Vault
7/28/2020, 11:38:45 AM	Crash Consistent	Vault
7/27/2020, 11:37:30 AM	Crash Consistent	Vault

When you back up data in Azure, you store that data in an Azure resource called a Recovery Services vault. The Recovery Services vault resource is available from the Settings menu of most Azure services. The benefit of having the Recovery Services vault integrated into the Settings menu of most Azure services is the ease of backing up data.

Here are the steps when you backup an Azure virtual machine:

- Create a Recovery Services vault
- Define a backup policy
- Apply the backup policy to protect multiple virtual machines

Hence, the correct answer is: **Recovery Services Vault.**

Backup policy is incorrect because you need to create a Recovery Services vault first. A backup policy is a schedule for how often and when recovery points are taken. A policy also includes the retention range for the recovery points.

Microsoft Azure Backup Server is incorrect. Microsoft Azure Backup Server (MABS) is a server product that can be used to back up on-premises physical servers, VMs, and apps running on them. The prerequisite of deploying a Microsoft Azure Backup Server is to have a Recovery Services Vault.

Recovery Plan is incorrect. A recovery plan gathers machines into recovery groups for the purpose of failover. A recovery plan helps you to define a systematic recovery process, by creating small independent units that you can failover. A unit typically represents an app in your environment. The requirement is to implement a backup solution not a disaster recovery solution.

References:

https://docs.microsoft.com/en-us/azure/backup/backup-overview

https://docs.microsoft.com/en-us/azure/backup/tutorial-backup-vm-at-scale

Check out this Azure Virtual Machines Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-machines/

7 Your company Azure Subscription contains the following resources:

Name	Type	Region
RSV1	Recovery Services vault	East Asia
VM1	Virtual Machine	West Asia
RSV2	Recovery Services vault	Central US
SA1	Storage Account	East Asia
VM2	Virtual Machine	Central US

You have created a file share named FS1 and a blob container named BC1.

Which of the following resources can be backed up in the Recovery Services vaults?

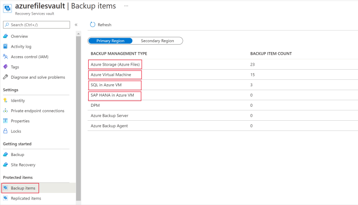
Select the correct answer from the drop-down list of options. Each correct selection is worth one point.

(view) 2 0 2 00:00:00

Backups can be performed using RSV1 FS1 and BC1 (FS1)
Choices: FS1 and BC1, SA1, FS1 and BC1, FS1, BC1

Backups can be performed using RSV2 VM1, FS1, BC1, and VM2 (VM2)
Choices: VM1 and VM2, VM2, VM1, SA1, and VM2, VM1, FS1, BC1, and VM2

A **Recovery Services vault** is a storage entity in Azure that houses data. The data is typically copies of data or configuration information for virtual machines (VMs), workloads, servers, or workstations. You can use Recovery Services vaults to hold backup data for various Azure services such as IaaS VMs (Linux or Windows) and Azure SQL databases. Recovery Services vaults support System Center DPM, Windows Server, Azure Backup Server, and more. Recovery Services vaults make it easy to organize your backup data while minimizing management overhead.



azurefilesvault | Backup items
Recovery Services vault

BACKUP MANAGEMENT TYPE	BACKUP ITEM COUNT
Azure Storage (Azure Files)	23
Azure Virtual Machine	15
SQL in Azure VM	3
SAP HANA in Azure VM	0
DPM	0
Azure Backup Server	0
Azure Backup Agent	0

In this scenario, you need to identify which resources can be backed up by RSV1 and RSV2. The first thing that you need to take a look at is the location or region of the resource. Since you can only backup using RSV if the resource and vault are in the same location.

You can only use Recovery Services vaults to hold backup data for various Azure services such as IaaS VMs (Linux or Windows) and SQL Server in Azure VMs. After knowing which resources can be backed up using RSV, the remaining resources would be VM and File Share.

Therefore, the correct answers are:

- **Backups can be performed using RSV1 = FS1**
- **Backups can be performed using RSV2 = VM2**

References:

https://learn.microsoft.com/en-us/azure/backup/backup-azure-recovery-services-vault-overview

https://learn.microsoft.com/en-us/azure/backup/backup-afs

https://learn.microsoft.com/en-us/azure/backup/backup-azure-arm-vms-prepare

Check out this Azure Virtual Machine Cheat Sheet:

https://tutorialsdojo.com/azure-virtual-machines/

9 You are responsible for overseeing the tutorialsdojo Azure web application.

It has come to your attention that users are frequently facing HTTP 500 errors during their interactions with the application.

To address this, you must ensure that the developers of tutorialsdojo app are provided with real-time and detailed visibility into these connection errors.

What would be your primary action?

(view) 1 0 1 00:00:00

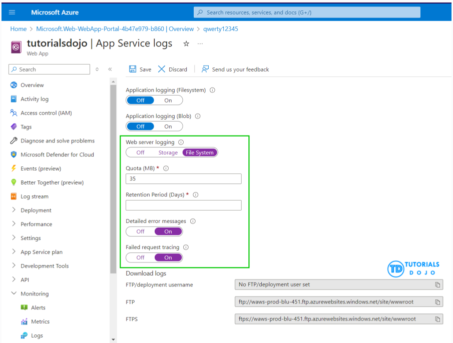
○ Create an Azure Monitor alert rule.

○ Establish a Service Health alert in Azure Monitor

○ Set up an Azure Monitor workbook.

○ Activate the Web server logging feature.

Web server logging is a feature provided by Azure that records raw HTTP request data in the W3C extended log file format. Each log message includes data such as the HTTP method, resource URI, client IP, client port, user agent, response code, and so on. This feature is particularly useful for monitoring and diagnosing issues related to web traffic and user requests.



tutorialsdojo | App Service logs

Application logging (Filesystem): Off

Application logging (Blob): Off

Web server logging: File System

Quota (MB) *: 35

Retention Period (Days) *:

Detailed error messages: On

Failed request tracing: On

Download logs

FTP/deployment username: No FTP/deployment user set

FTP: ftp://waws-prod-blu-451.ftp.azurewebsites.windows.net/site/wwwroot

FTPS: ftps://waws-prod-blu-451.ftp.azurewebsites.windows.net/site/wwwroot

In the context of the given question, users are frequently encountering HTTP 500 errors when interacting with the `tutorialsdojo` application. HTTP 500 errors are server-side errors, which means they occur when the server fails to fulfill a request due to an issue with the server itself. These errors can be caused by various issues, such as server overload, server malfunction, or programming errors.

By activating the Web server logging feature for `tutorialsdojo`, you would be able to capture raw HTTP request data, which includes crucial details about these HTTP 500 errors. This would provide the developers of tutorialsdojo with real-time access to connection errors, allowing them to diagnose and address the issues causing the HTTP 500 errors.

Hence, the correct answer is: **Activate the Web server logging feature.**

The option that says: **Create an Azure Monitor alert rule** is incorrect because it is primarily used for triggering alerts based on specific conditions, such as error thresholds or metrics. However, it does not capture detailed logs of HTTP 500 errors.

The option that says: **Establish a Service Health alert in Azure Monitor** is incorrect because Service Health alerts are about the health of Azure services rather than specific application-level errors like HTTP 500 errors.

The option that says: **Set up an Azure Monitor workbook** is incorrect because while Azure Monitor Workbooks can be useful for analyzing and visualizing data, they simply don't provide real-time access to connection errors.

References:

https://learn.microsoft.com/en-us/azure/app-service/troubleshoot-diagnostic-logs

https://techcommunity.microsoft.com/t5/apps-on-azure-blog/azure-app-service-logging-how-to-monitor-your-web-apps-in-real/ba-p/3800390

Check out this Azure App Service Cheat Sheet:

https://tutorialsdojo.com/azure-app-service/

10 You are an administrator managing your organization's Azure environment. One of your responsibilities includes monitoring and managing Azure Functions, which are used extensively for processing events from Azure Event Hubs.

Recently, users have reported delays in event processing, and you need to identify the cause. Upon investigating, you notice that the function app has a high number of unprocessed events. Your organization also uses the Azure Performance Diagnostics agent to collect performance data from Azure virtual machines.

Which metric should you use to monitor the unprocessed events?

(view) 1 0 1 00:00:00

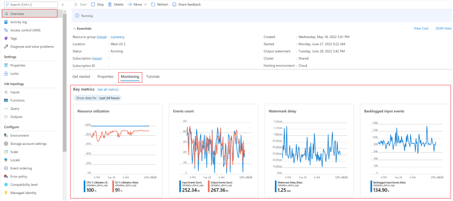
○ Function Execution Errors

○ Watermark Delay

○ Out-of-Order Events

○ Backlogged Input Events

Backlogged Input Events is a critical metric for identifying delays in event processing within an Azure Function. This metric explicitly measures the number of events waiting to be processed by the function. A high number of backlogged input events indicates that the function cannot keep up with the incoming data from the Event Hub, leading to a queue of unprocessed events. Monitoring this metric allows administrators to pinpoint the exact cause of processing delays and take appropriate actions, such as scaling the function app or optimizing the code to improve throughput.



Running

Essentials

Resource group (move) : xujxasarg

Location : West US 2

Status : Running

Subscription (move) :

Subscription ID :

Created : Wednesday, May 18, 2022 3:41 PM

Started : Monday, June 27, 2022 9:22 AM

Output watermark : Tuesday, June 28, 2022 3:42 PM

Cluster : Shared

Hosting environment : Cloud

Show data for : Last 24 hours

Administrators can effectively manage and mitigate issues related to event processing delays by focusing on the Backlogged Input Events metric. Unlike other metrics, such as Late Input Events or Watermark Delay, which provide information on the timing and order of event arrivals, Backlogged Input Events directly highlight the accumulation of unprocessed data. This enables more targeted troubleshooting and resource allocation, ensuring that the Azure Function can handle the event load efficiently and maintain optimal performance.

Hence, the correct answer is: **Backlogged Input Events.**

Function Execution Errors is incorrect. This metric primarily indicates the number of times your function code fails to execute successfully due to exceptions or timeouts, but it does not provide information about the backlog of unprocessed events waiting to be pulled from Event Hubs.

Watermark Delay is incorrect because this metric only measures the delay between the event time (when the event was generated) and the processing time (when the event is processed by the function). This metric can highlight latency issues but does not specifically indicate the number of events that are pending processing.

Out-of-Order Events is incorrect because this option tracks events that arrive out of their intended sequence. While important for ensuring the correct order of event processing, it does not provide any information about the backlog of events. It focuses on the sequence and order rather than the quantity of events waiting to be processed.

References:

https://learn.microsoft.com/en-us/azure/stream-analytics/stream-analytics-time-handling

https://learn.microsoft.com/en-us/azure/stream-analytics/stream-analytics-job-metrics

Check out this Azure Big Data Cheat Sheet:

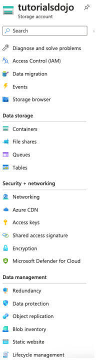
https://tutorialsdojo.com/azure-big-data/

12 You have an Azure storage named tutorialsdojo for hosting static assets. You have a user named Manila that uploads images to a blob container in tutorialsdojo.

You need to ensure that you satisfy the following requirements:

- Requirement 1: Only allow access from the specific public internet IP address of Manila.

Requirement 2: Data accidentally deleted must be recoverable 14 days after deletion.

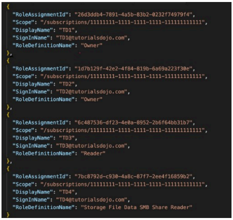


Which two storage account features should you use to satisfy requirements?

Select the correct answer from the drop-down list of options. Each correct selection is worth one point.

(view) 2 0 2 00:00:00

13 You have a role assignment file below:



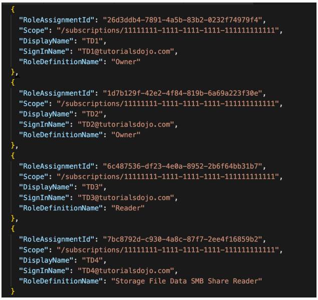
```
{
  "RoleAssignmentId": "26d3ddb4-7891-4a5b-83b2-0232f74979f4",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD1",
  "SignInName": "TD1@tutorialsdojo.com",
  "RoleDefinitionName": "Owner"
},
{
  "RoleAssignmentId": "1d7b129f-42e2-4f84-819b-6a69a223f30e",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD2",
  "SignInName": "TD2@tutorialsdojo.com",
  "RoleDefinitionName": "Owner"
},
{
  "RoleAssignmentId": "6c487536-df23-4e0a-8952-2b6f64bb31b7",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD3",
  "SignInName": "TD3@tutorialsdojo.com",
  "RoleDefinitionName": "Reader"
},
{
  "RoleAssignmentId": "7bc8792d-c930-4a8c-87f7-2ee4f16859b2",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD4",
  "SignInName": "TD4@tutorialsdojo.com",
  "RoleDefinitionName": "Storage File Data SMB Share Reader"
}
```

For each of the following items, choose **Yes** if the statement is true or choose **No** if the statement is false. Take note that each correct item is worth one point.

(view) 3 0 3 00:00:00

Questions	Yes	No
TD1 can deploy virtual machines.	○	○
TD3 can delete resources.	○	○
TD4 can provision file shares.	○	○

Azure role-based access control (Azure RBAC) is a system that provides fine-grained access management of Azure resources. Using Azure RBAC, you can segregate duties within your team and grant just the right access to users that they need to perform their jobs.



```
{
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  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD1",
  "SignInName": "TD1@tutorialsdojo.com",
  "RoleDefinitionName": "Owner"
},
{
  "RoleAssignmentId": "1d7b129f-42e2-4f84-819b-6a69a223f30e",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD2",
  "SignInName": "TD2@tutorialsdojo.com",
  "RoleDefinitionName": "Owner"
},
{
  "RoleAssignmentId": "6c487536-df23-4e0a-8952-2b6f64bb31b7",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD3",
  "SignInName": "TD3@tutorialsdojo.com",
  "RoleDefinitionName": "Reader"
},
{
  "RoleAssignmentId": "7bc8792d-c930-4a8c-87f7-2ee4f16859b2",
  "Scope": "/subscriptions/11111111-1111-1111-1111-111111111111",
  "DisplayName": "TD4",
  "SignInName": "TD4@tutorialsdojo.com",
  "RoleDefinitionName": "Storage File Data SMB Share Reader"
}
```

Role assignments enable you to grant a principal (such as a user, a group, a managed identity, or a service principal) access to a specific Azure resource. This article describes the details of role assignments.

Access to Azure resources is granted by creating a role assignment, and access is revoked by removing a role assignment.

A role assignment has several components, including:

- The principal or who is assigned the role.
- The role that they're assigned.
- The scope at which the role is assigned.
- The name of the role assignment and a description that helps you to explain why the role has been assigned.

Based on the role assignment file given:

- `TD1` and `TD2` has the Owner role which means it has full access to manage all resources, including the ability to assign roles in Azure RBAC.
- `TD3` has the Reader role, which means it can only view all the resources but is not allowed to make any modifications.
- `TD4` has the Storage File Data SMB Share Reader, which allows for read access on files/directories in Azure file shares.

Hence, the correct statement is: **`TD1` can deploy virtual machines.**

The option that says: **`TD3` can delete resources** is incorrect. Since TD3 has the reader role, it can only view resources. But it won't be able to delete resources.

The option that says: **`TD4` can provision file shares** is incorrect. Since TD4 has an SMB share reader, it can only view files and directories in your file share. It won't be able to provision new file shares.

References:

https://learn.microsoft.com/en-us/azure/role-based-access-control/built-in-roles

https://learn.microsoft.com/en-us/azure/role-based-access-control/role-assignments

Check out this Azure RBAC Cheat Sheet:

https://tutorialsdojo.com/azure-role-based-access-control-rbac/

Microsoft Entra ID vs Role-Based Access Control (RBAC):

https://tutorialsdojo.com/microsoft-entra-id-vs-role-based-access-control-rbac/

Sub-Total:	20	7 (35%)	13 (65%)	00:00:03	35%
Total	70	35 (50%)	35 (50%)	00:00:03	50%

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